PhaseAtlas: From PDE Animations to Reproducible Numerical Evidence

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Abstract

PhaseAtlas addresses a recurring problem in computational PDE research: visual accessibility is often separated from the evidence needed to assess a result. We present an independently implemented browser workbench that connects model editing, animated simulation, controlled refinement, travelling-wave diagnostics and finite-time parameter maps. The system retains configurations, boundary conditions and trajectories with each experiment and recomputes imported diagnostics from numerical arrays. A Robin heat-equation eigenmode exhibits observed spatial order 1.99995 in a fixed-time-step experiment. A boundary-cooling study demonstrates a practical comparison of diffusivity and heat exchange; a fourth-order front from the author's MAE03 research supplies a mathematical case study. The contribution is an integrated, inspectable software workflow. Numerical observations, same-mesh reference agreement and mathematical theorems are kept distinct.

1. Research problem and intended users

A smooth animation is a useful starting point for understanding a PDE, but it is not evidence that the discretization resolves that equation. A changed boundary condition can change the scientific problem. Grid error can mimic growth. Translational drift can inflate the difference between two otherwise similar wave profiles. The researcher must ask which quantity was measured, what was held fixed, and whether another implementation reproduces the observation.

The primary users are PDE researchers, scientific-computing developers and research instructors. PhaseAtlas is intended to reduce the repeated work of building a small experimental interface, adding diagnostics and packaging a reproducible comparison. No user study has yet measured a time saving. The current evidence supports technical feasibility and reusable numerical workflows.

2. Position relative to existing tools

VisualPDE already provides accessible interactive PDE simulation and published validation examples [1]. Dedalus provides a substantial framework for solving equation-based spectral problems [2]. PhaseAtlas does not claim to invent browser simulation, refinement studies or parameter scans, and no performance or accuracy superiority over those tools is established. Its present emphasis is the combination of an inspectable experiment record, wave-specific diagnostics and metric-defined parameter maps in one small browser workflow. The implementation used here was developed independently; it does not incorporate VisualPDE source code or assets.

Workflow

Define	Compute	Inspect	Exchange
Equations, initial data, domain and boundary rows	Workers retain numerical time histories	Resolution, balance, motion, shape and parameter response	Model, arrays, settings and method identifiers

3. Numerical scope and implementation

The Studio supports one to three fields on 1D intervals or 2D rectangles, subject to backend restrictions. A restricted expression parser accepts mathematical expressions rather than executable JavaScript. The browser uses CPU Float64 arithmetic and worker execution. The server hosts assets and health endpoints; it is not a remote scientific-computing cluster.

Backend	Supported setting	Principal restrictions
Periodic	1D/2D Fourier spatial derivatives; ETDRK4 integration	Extracted constant diagonal linear operators; explicit supported remainder; resolution limits
Second-order bounded	1D/2D cell-centered differences and SBDF2	Constant nonnegative diagonal diffusion; Dirichlet, Neumann or passive Robin data
Fourth-order bounded	1D cell-centered stencils and SBDF2	Negative constant leading fourth coefficient per field; two admissible linear rows per end

For the second-order backend, the stiff diagonal diffusion is implicit and supported remaining terms are explicit. Nonlinear advection is not shock capturing. In the fourth-order backend, constant same-field linear derivative terms are implicit; boundary traces are reconstructed from nearby cells and two specified linear rows. A singular reconstruction or failed solve is rejected. An admissible discrete boundary matrix is not, by itself, a proof that an arbitrary continuum boundary problem is well posed.

3.1 Boundary values are part of the experiment

The Robin convention is $\alpha \cdot u + \beta \cdot \partial_n u = g$, where ∂_n is the outward normal derivative. At the left endpoint $\partial_n = -\partial_x$; at the right $\partial_n = \partial_x$. The current Robin backend requires finite nonnegative constant/parameter α and β, not both zero; g may depend on position, time and parameters. These restrictions support the implemented passive diffusion solve. General active Robin feedback is not claimed.

At a cell center U a distance $h/2$ from a boundary, the reconstructed normal derivative is $q = (g - \alpha \cdot U)/(\beta + \alpha \cdot h/2)$. The resulting face input is included in the discrete diffusion operator. Taking $\alpha = 0$ or $\beta = 0$ recovers the supported Neumann or Dirichlet closure after scaling. The fourth-order editor separately allows linear combinations of u, $\partial_n u$, $\partial_n^2 u$ and $\partial_n^3 u$, with two independent rows per end.

3.2 Evidence cannot be inferred from a small residual alone

A small discrete boundary residual records enforcement of the chosen boundary rows. A small algebraic balance closure records consistency of assembled terms. Neither measures error against the continuous PDE. PhaseAtlas therefore exposes exact-solution comparisons, controlled refinement and an independent Python implementation alongside these internal checks.

4. Case study I: verified boundary cooling

The verification problem is $u_t = v \cdot u_{xx}$ on [-1,1], with v = 0.1, initial data cos(x), and Robin data tan(1) · u + ∂_n u = 0 at both ends. Its exact solution is exp(-0.1 · t) · cos(x). Three independent runs use N = 16, 32 and 64 cell centers, final time 0.2 and time step 0.0002. Space is refined while time step, equation and boundary data are held fixed.

Cells N	Final RMS error to exact solution
16	4.8629086e-5
32	1.2504878e-5
64	3.1475601e-6

Adjacent-grid differences are compared on a common coarse grid. Their ratio gives p = $\log(D_{01}/D_{12})/\log(2)$ = 1.999945875. The exact errors and observed order are consistent with second-order behavior in this tested regime. They do not certify every equation, boundary combination or parameter setting accepted by the editor.

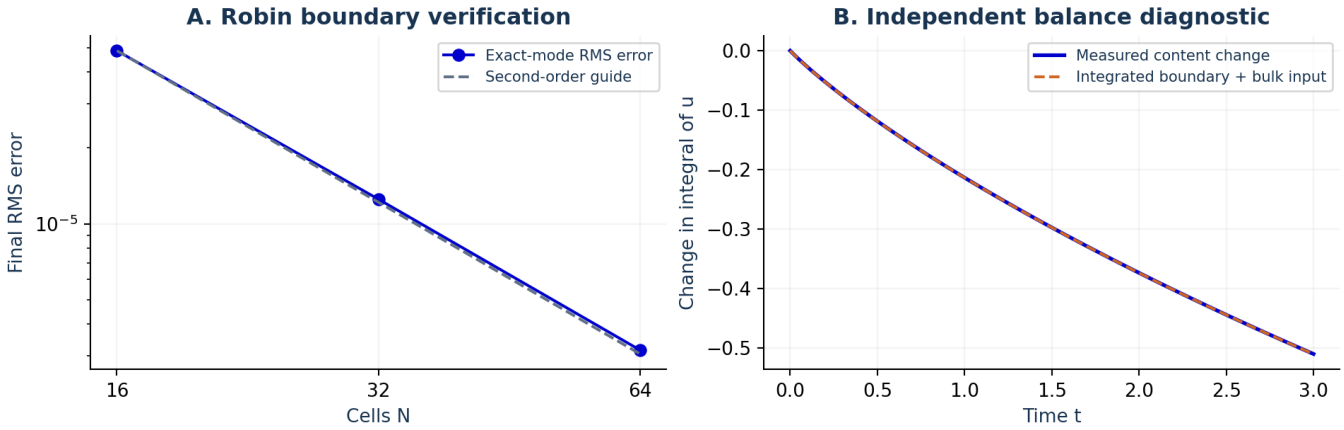


Figure 1. Left: exact-mode error under spatial refinement. Right: a separate cooling experiment, whose thermal-content change is compared with integrated input. Both plots use executed numerical output.

4.1 A physically interpretable application

A second experiment starts from u = 1 with surrounding value 0.2. It uses $u_t = v \cdot u_{xx}$ and Bi · u + ∂_n u = Bi · 0.2, where Bi is a dimensionless boundary-exchange coefficient. With v = 0.1, Bi = 2, N = 64, time step 0.001 and final time 3, the computed mean is 0.74485847. The initial uniform field is incompatible with the suddenly applied cooling boundary at t = 0; this is explicitly treated as a start-up transient.

For constant volumetric heat capacity, the integral of temperature is proportional to thermal content. The audit compares its change with the saved-time integral of boundary and bulk input. With 121 saved frames, the maximum absolute discrepancy is 8.711e-5 in dimensionless integral units. This combines numerical error and trapezoidal quadrature of saved output; it is not a calibrated energy-error tolerance. Physical prediction would require a specified material, dimensions, units, coefficients and experimental validation.

5. From one run to a useful parameter question

A researcher may ask how much of a hot sample remains above ambient after a fixed time, and whether internal diffusion or surface exchange limits cooling. The workbench evaluates a user-selected expression and spatial statistic, then applies a temporal reduction such as final value, time average, maximum or first threshold event. Each map cell retains the experiment used to compute that quantity.

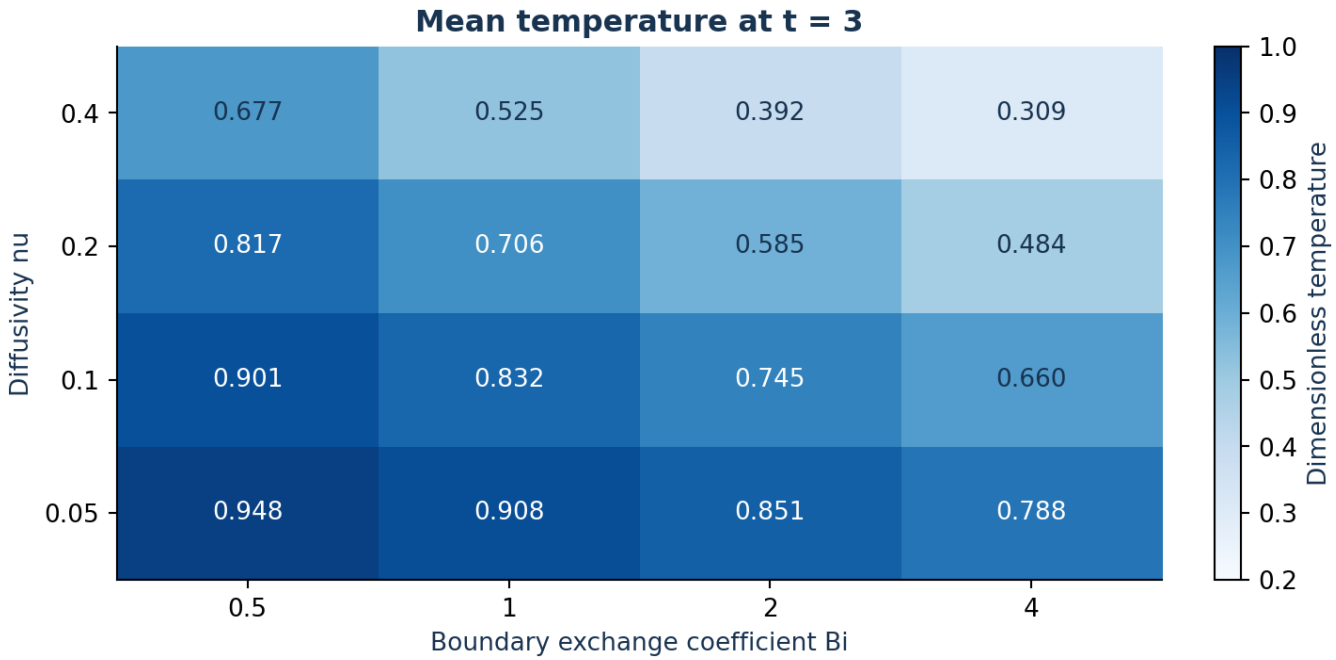


Figure 2. Final mean temperature at t = 3 for a 4-by-4 scan of diffusivity and boundary exchange. Every cell is an independent initial-value run from u = 1, using N = 32 and time step 0.001. Lower values represent more cooling in this model.

The sampled final means range from 0.30895041 to 0.94776467. The map exposes a response that a single animation does not: changing either diffusion or wall exchange changes the finite-time average, and the response depends on both parameters. These are finite-grid model outputs, not a newly discovered physical law. A decision near a threshold must be checked with finer spatial grids, smaller time steps and denser output sampling.

Missing evidence stays missing

Canceled and failed cells remain visible. An unobserved threshold event is missing rather than zero. An ambiguous wave crossing does not produce a fabricated speed. Finite-difference parameter sensitivities are reported as local secants, with one-sided treatment at sampled edges and unavailable values when necessary neighbors are missing. The interface does not relabel this finite-time scan as continuation of stationary solutions or a bifurcation diagram.

Practical reuse

The exported model can be adapted to another supported diffusion or reaction problem; trajectories and metric definitions can be inspected without rerunning it. A collaborator can replay a chosen cell, change the observation window and repeat the numerical study. These capabilities support model exploration and teaching. Application-specific clinical, industrial or safety decisions are outside the validated scope of this demonstration.

6. Case study II: a fourth-order travelling front

The mathematical case comes from the author's earlier MAE03 investigation and the travelling-wave literature [3,4]. The equation is $u_t = \gamma \cdot u_{xxxx} - \beta \cdot u_{xxx} + \alpha \cdot u_{xx} - \partial_x(a \cdot u^2 + 2 \cdot b \cdot u^3/3)$. The reference parameters are $\gamma = -1$, $\beta = -15/2$, $\alpha = -71/4$, $a = 0$, $b = 315/16$ and speed $c = 105/8$. The profile $v(\xi) = (1 + \exp(-\xi))^{-3/2}$, $\xi = x - c \cdot t$, connects 0 to 1. Its strictly positive base makes the half-integer power real throughout the line.

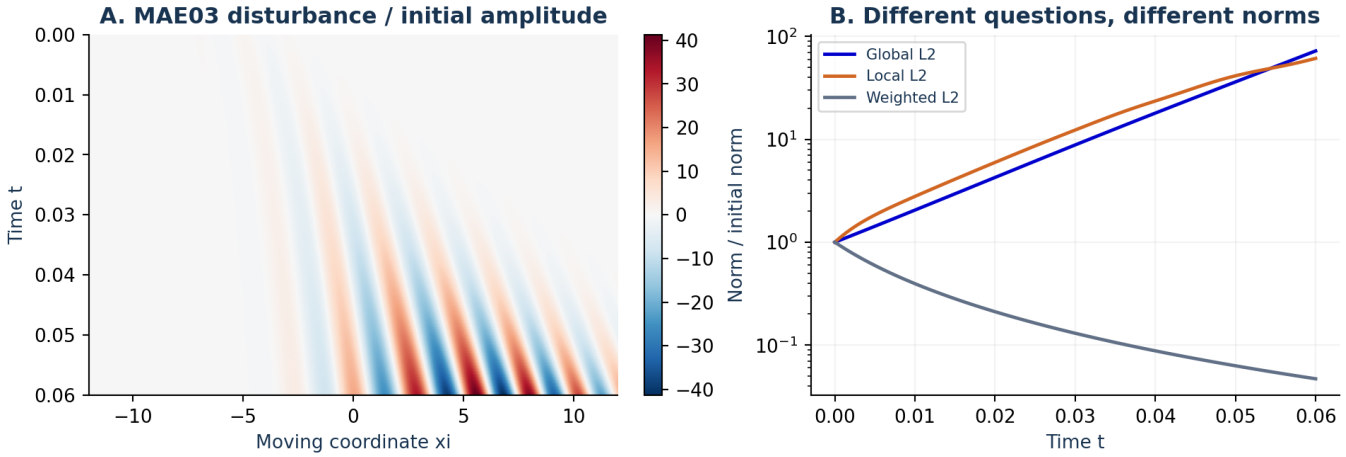


Figure 3. A localized disturbance in the original front laboratory. Left: w/0.001 on one fixed color scale. Right: global, local and exponentially weighted finite-domain L2 norms, each divided by its own initial value. These are distinct observations, not a stability classification.

The plotted nonlinear perturbation uses N = 2049, interval [-24,24], final time 0.06, time step 0.000025, amplitude 0.001 and 121 saved frames. The original laboratory evolves a perturbation equation with homogeneous clamped perturbation conditions. Its zero-perturbation state is well balanced by construction and is not an independent test of solver accuracy.

Motion, deformation and boundaries

In the general workbench, threshold tracking measures motion, while windowed phase alignment helps separate displacement from shape change. Global, local and weighted raw-difference norms answer different questions. The supplied reference formula need not be an exact solution of a perturbed experiment. Weighted decay, short-time growth or a finite-domain trace alone does not establish nonlinear orbital or convective stability.

The newer general fourth-order Studio evolves the full field with user-specified endpoint rows. It is a different numerical path from the original perturbation laboratory and can drift even at zero added perturbation. Users must first measure that baseline drift, then vary grid, time step and domain. Prescribed finite boundaries are not absorbing whole-line conditions.

7. Reproducibility and evidence status

The case-data generator writes complete trajectories, a numerical summary and SHA-256 hashes of the numerical source files. These hashes identify the files used; they do not authenticate an arbitrary imported trajectory. The evidence viewer validates structure and recomputes comparisons from arrays rather than trusting saved conclusions. Study import restores recipes and settings without automatically accepting numerical claims.

The 18 October snapshot passes 90 mathematical and numerical tests. Coverage includes parser rejection, Fourier signs, Dirichlet/Neumann/Robin closures, fourth-order boundary rows, manufactured solutions, refinement, conservation behavior, missing-data handling and wave diagnostics. Tests provide targeted regression evidence, not universal certification. The latest preceding browser QA also exercised 30 workbench/Robin scenarios; presentation changes are checked separately.

An independent NumPy/SciPy BDF run for the Robin mode at N = 32, final time 0.3 and browser time step 0.0002 gives a final relative L^2 difference of 2.174e-10. The Python and browser implementations use the same spatial closure, so this is a same-mesh implementation/time-integration comparison. The exact-mode refinement on page 3 addresses spatial error separately. The reference input and report are included in the package.

8. Contribution, limitations and next evaluation

The current contribution is the connected scientific-software workflow and its explicit experiment records. The project does not claim a universal PDE solver, numerical superiority over established software, autonomous theorem discovery or a new mathematical theorem. General geometries, unrestricted variable-coefficient operators, 2D nonperiodic fourth-order problems, GPU acceleration and remote job execution remain outside this snapshot.

The next evaluation should ask researchers to reproduce and modify a benchmark, compare results across implementations and document failures. A useful study would measure task completion, interpretation mistakes and reproducibility, rather than count interface features. A later mathematical paper requires a distinct new result and proof. This software article does not substitute for that work.

Provenance and event status

Author: Niu Te, Imperial College London. Development was AI-assisted, including coding, tests and documentation. The scientific computation does not call an external LLM at runtime. Earlier mathematical work and this dated software snapshot precede the October 21-22 event; only separately recorded event-period changes should be described as hackathon contributions. This manuscript is a project report prepared before the event, not a peer-reviewed paper.

References and access

[1] VisualPDE, Validating VisualPDE. visualpde.com/numerical-methods/validating-VisualPDE.html
[2] Dedalus Project, About. dedalus-project.org/about/
[3] Y. Hu and Q. Liu (2019), On traveling wave solutions of a class of KdV-Burgers-Kuramoto type equations. AIMS Mathematics 4, 1450-1465. doi:10.3934/math.2019.5.1450
[4] M. A. Johnson, G. Lyng and C. Smith, On the Dynamics of Traveling Fronts Arising in Nanoscale Pattern Formation. arXiv:1810.02233
Project source: github.com/NIU-55383/PhaseAtlas. Numerical case files and generation scripts accompany this report. Web references checked 18 October 2026.